# ⭐ MONTH 2 – HIGH-WEIGHTAGE CLASS 12 CHAPTERS

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# 📘 PHYSICS (Class 12)

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### 1. Magnetism & Moving Charges

**Subtopics:**

1. Magnetic Force on Moving Charge
2. Motion in Magnetic Field (Cyclotron, Radius, Frequency)
3. Magnetic Force on Current-Carrying Conductor
4. Biot–Savart Law
5. Magnetic Field due to
    - Straight Current-Carrying Wire
    - Circular Loop
    - Solenoid
6. Ampere's Circuital Law
7. Magnetic Materials (Dia, Para, Ferro)
8. Hysteresis Curve
9. Magnetic Dipole
10. Torque on Current Loop
11. Bar Magnet as Equivalent Solenoid

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### 2. Electromagnetic Induction (EMI)

**Subtopics:**

1. Magnetic Flux
2. Faraday's Laws of EMI
3. Lenz's Law
4. Induced EMF
5. Eddy Currents
6. Self-Induction
7. Mutual Induction
8. Energy Stored in Inductor

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### 3. Alternating Current (AC)

### Subtopics:

1. AC Voltage & Current
2. RMS & Peak Values
3. Phase Difference
4. Impedance & Reactance
   - Capacitive Reactance
   - Inductive Reactance
5. LCR Series Circuit
6. Resonance
7. Power in AC Circuits
8. Power Factor
9. Transformers

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# 📗 CHEMISTRY (Class 12)

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## 1. Chemical Kinetics

### Subtopics:

1. Rate of Reaction
2. Factors Affecting Rate
3. Rate Law & Rate Constant
4. Order and Molecularity
5. Integrated Rate Laws
   - Zero Order
   - First Order
6. Half-life
7. Temperature Dependence – Arrhenius Equation
8. Collision Theory
9. Catalysis & Activation Energy

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## 2. Surface Chemistry

### Subtopics:

1. Adsorption (Physisorption & Chemisorption)
2. Adsorption Isotherms (Freundlich, Langmuir)
3. Catalysis
4. Colloids

- Types
- Preparation
- Purification
- Properties (Tyndall effect, coagulation)

5. Emulsions

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### 3. The p-Block Elements (Groups 13–18)

**Subtopics:**

1. Group Trends
2. Important Compounds:
   - Boron compounds, Borax, Boric Acid
   - Aluminium compounds ($AlCl_3$ etc.)
   - Carbon & Silicates
   - Nitrogen & Phosphorus oxides and acids
   - Sulphur compounds ($SO_2$, $SO_3$, $H_2SO_4$)
3. Halogens ($Cl_2$, $Br_2$, $I_2$ – Properties)
4. Noble Gases & Compounds ($XeF_2$, $XeF_4$, $XeF_6$)
5. Structures of Important Molecules

*(You don't memorize everything; focus on NCERT-most important)**

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## 📘 MATHEMATICS (Class 12)

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### 1. Continuity

**Subtopics:**

1. Definition of Continuity
2. Left- & Right-hand Limits
3. Algebra of Continuous Functions
4. Continuity of Standard Functions
5. Graph-based continuity
6. Continuity at a Point & in an Interval

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### 2. Differentiability

**Subtopics:**

1. Definition of Derivative

2. Relation between Continuity & Differentiability
3. Non-differentiable Graphs (sharp points, cusp, discontinuity)
4. Derivatives of Standard Functions
5. Chain Rule
6. Logarithmic Differentiation

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### 3. Application of Derivatives (AOD)

**Subtopics:**

1. Rate of Change of Quantities
2. Increasing/Decreasing Functions
3. Maxima & Minima
4. Tangents & Normals
5. Approximation ($\Delta y \approx dy$)
6. Rolle's Theorem & Lagrange's Mean Value Theorem (conceptual)
7. Marginal Cost / Revenue (for JEE Main sometimes asked)

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## 🔥 1-MONTH (4-WEEK) PLAN: Month 2

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### Week 1

**Physics: Magnetism**

**Chemistry: Chemical Kinetics**

**Math: Continuity**

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### Week 2

**Physics: EMI**

**Chemistry: Surface Chemistry**

**Math: Differentiability**

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### Week 3

**Physics: AC**

**Chemistry: p-Block**

**Math: Application of Derivatives**

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## Week 4

- Complete p-Block revision
- Solve NCERT + PYQs (JEE + Boards)
- Mixed mock tests (PCM)
- Revise formulas & error-prone topics